INCH-POUND
MIL-DTL-24643/57E
22 August 2002
SUPERSEDING
MIL-C-24643/57D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2UW-42
(INCLUDING VARIATION LS2UWS-42)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

This cable can be manufactured in two variations: LS2UW-42 and LS2UWS-42 (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG size 26, 7 strands.
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.010 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Telephone identification applied by method 3.
Fourth -	Two insulated conductors twisted together with a nominal lay of 1-1/2 inches to form a pair.
Fifth -	Forty-two pairs cabled together with a maximum lay of 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Sixth -	Binder tape applied helically with overlap.
Seventh -	Braided shield of tin coated copper.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

Construction variant for type LS2UW-42

Eighth - A separator tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.065 inch. Cable overall diameter shall be 0.750 minimum, 0.790 maximum inch.

Construction variant for type LS2UWS-42

Eighth - Spirally wrapped 0.002 inch thick polyethylene (polyester film) separator tape with 25 percent minimum overlap.
Ninth - Braided shield of tin coated copper.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Minimum average wall thickness 0.065 inch. Cable overall diameter shall be 0.785 minimum, 0.825 maximum inch.

Notes: 1. Variant LS2UW-42 military part number - M24643/57-01UO.
2. Variant LS2UWS-42 military part number - M24643/57-03UD.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	45.19
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Megohms -1000 feet minimum	
Conductor to conductor	500
Conductor to shield	500
Shield to shield (LS2UWS-42)	100
Jacket flaws	No failure
Component and shield continuity	No failure

Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	115 ±15
Watertightness - leakage, in^3, maximum	1.0

Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at minus 30 ± 2°C, over a	
9 inch diameter mandrel	No damage
Drip - 95 ± 1 °C	Zero
Bending endurance - at 25 ± 2°C, cycles minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160

Group C:

Physicals - (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
- Hot oil immersion
  - Tensile strength - percent of unaged, minimum 50
  - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125
- Cable sealant removability ..... No failure
- Shield conformance to material, construction and coverage ..... No failure
- Surface transfer impedance (LS2UWS only)
  - Milliohms per meter, maximum 700
  - EMP response time - db, minimum 60

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

- Aging and compatibility (cable) (125 ± 5°C) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
- Halogen content - percent, maximum
  - Jacket 0.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.

Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-055)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:18Z
Check the source to verify that this is the current version before use.